Jerry W. Xuan

51 Pegasi b Fellow
University of California, Santa Barbara

https://wenhaoxuan.github.io/
jxuan@ucsb.edu

Papers: 81 total (refereed) · **First author:** 13 · **Second or third-author:** 23 · **Student-led:** 5
Citations: 1,600 · ***h*-index:** 23 · SciX · ORCID

Education

Ph.D. in Astrophysics 2020–2025
California Institute of Technology
"Probing the Origins of Directly Imaged Planets and Brown Dwarfs: From Atmospheric Compositions to Binarity"

M.Phil. in Astronomy 2019–2020
University of Cambridge
"Constraining 3D orbital architectures in systems with giant planets"

B.A. in Physics 2015–2019
Pomona College (Cum Laude, Downing Scholar)

Academic Appointments

51 Pegasi b Fellow, University of California, Santa Barbara — Oct. 2026–Present
51 Pegasi b Fellow, University of California, Los Angeles — Oct. 2025–Sept. 2026
Postdoctoral Scholar, California Institute of Technology — 2025 (7 months)
Keck Visiting Scholar, W. M. Keck Observatory — 2022 (4 months)

Research Interests

- Planet formation and evolution; chemical abundances in exoplanet and brown dwarf atmospheres.
- Planetary system orbital architectures, dynamical evolution, and demographics.
- High-resolution spectroscopy, high-contrast imaging, atmospheric modeling, and absolute astrometry.

Grants and Observing Programs

JWST: PI/Co-PI of five programs, 117 hours across Cycles 2–5

Total awarded grants: $780k

PI GO 11023 (32.0 hours, **$243,632**) JWST Cycle 5
A NIRSpec Spectroscopic Search for Astrometric Giant Planet Candidates at Solar System-like Scales

Co-PI GO 9993 (44.0 hours, **$103,581**) JWST Cycle 5
Measuring the composition and circumplanetary environment of a gap-clearing protoplanet

PI GO 8714 (23.6 hours, **$159,288**) JWST Cycle 4
Combining isotopic and elemental abundances to unveil the formation and accretion history of a cold Jupiter

PI GO 5342 (13.2 hours, **$150,900**) JWST Cycle 3
Spectroscopic characterization of the lowest-mass imaged Jupiter analog

PI GO 3762 (3.8 hours, **$74,911**) JWST Cycle 2
Stringent tests of atmospheric and evolutionary models with a benchmark T dwarf companion

Additional JWST Programs as Co-I

GO 12182: 18.7 hours • GO 10963: 13.3 hours	Cycle 5
DDT 12628: 4.5 hours • DDT 12518: 12.9 hours • GO 8877: 25.9 hours	Cycle 4
GO 8063: 59.0 hours; atmospheric analysis lead	Cycle 4
GO 6362: 30.3 hours; atmospheric analysis co-lead • AR 6199	Cycle 3
GO 4982: 19.3 hours; atmospheric analysis lead (**$42,619**)	Cycle 3

ESO Very Large Telescope

Total of 22 hours as PI on VLTI/GRAVITY (equivalent to 9 nights)	P112, P117
9 hours as co-PI on VLTI/GRAVITY (equivalent to 4 nights)	FTC P117
Total of 35 hours as PI on VLT/CRIRES+	P112, P115
Over 100 hours as co-I on VLTI/GRAVITY	P114, P116, P117

Keck Observatory

Total of 6 nights as PI on KPIC, NIRSPEC, NIRC2	2025–2026
Total of 12 nights as Science PI on KPIC, NIRC2	2021–2026
Over 40 nights as co-I on KPIC, NIRC2	2018–2025

Gemini Observatory

20 hours as PI on IGRINS	2023

WIYN Observatory

4 nights as co-PI on NEID	2026B

Fellowships, Honors, and Awards

Fellowship Support: $600k

Heising-Simons Foundation 51 Pegasi b Fellow (**$450k**)	2025–Present
NASA Hubble Fellowship Program Sagan Fellow	2025 (Declined)
Future Investigators in NASA Earth and Space Science and Technology (**$150k**)	2023–2025
Phi Beta Kappa	2019
Sigma Xi	2019
Downing Scholarship (for M.Phil. at Cambridge)	2019
The Brackett Prize in Astronomy, Pomona College	2019
Tileston Physics Prize, Pomona College	2018

Collaborations, Service and Outreach

Keck/SCALES Science Team Member	2025–Present
GPI 2.0 Exoplanet Survey Team Member	2026–Present
VLTI ExoGRAVITY Team Member	2025–Present
Keck/KPIC Instrument Team Member	2020–2026
Keck AO Working Group & NGS/LGS Subgroup Member	2025–Present
ExSoCal Scientific Organizing Committee Member	2025
Referee for AAS Journals, A&A, MNRAS	2021–Present

Habitable Worlds Observatory Working Group; Spectral Processing Focus Group 2024–2025

Guest Lecture in Astro 51 at Pomona College 2025

Astronomy Outreach at Partial Solar Eclipse 2023

Public Talk, Keck Observatory Visiting Scholars Program, Waimea, HI 2022

Public Open Evenings at Institute of Astronomy, Cambridge 2020

Mentoring

Gavin Wang, undergraduate (JHU) → PhD student (Caltech) 2024–Present

Project: Chemical abundances of brown dwarf companions with high-resolution spectroscopy.

Publication: First-author ApJ paper (2026).

Aniket Sanghi, PhD student at Caltech (partly funded by GO 8714: PI Xuan) 2021–Present

Project: Direct imaging with Keck/NIRC2 and JWST imaging of ϵ Indi Ab.

Publications: Two first-author AJ papers (2024, 2026).

Marvin Morgan, PhD student at UCSB 2025–Present

Project: Precise orbits for stellar companions to hot Jupiters with NIRC2 and GRAVITY.

Publication: First-author AJ paper (accepted, 2026).

Carrie He, PhD student at UCLA (funded by GO 5342: PI Xuan) 2025–Present

Project: Connecting JWST spectroscopy of directly imaged planets to planet formation.

Publication: First-author paper in preparation.

Sage Santomenna, undergraduate (Pomona College) → PhD student (UNC Chapel Hill) 2025–Present

Project: Characterizing substellar binarity with VLT/CRIRES+ high-resolution spectroscopy.

Publication: First-author paper in preparation.

Samuel Whitebook, PhD student at Caltech 2024–Present

Project: Searching for substellar binaries with Keck/NIRSPEC high-resolution spectroscopy

Additional Mentoring

Aylin García Soto (Dartmouth) 2025–2026

AMP-UP mentoring for postdoctoral applications. Placement: Postdoctoral position at Boise State University.

Jacob Schimp (UIUC) 2024

Mentoring for graduate school applications. Placement: Graduate study at UT Austin.

Teaching Experience

Teaching Assistant, California Institute of Technology 2021

Bayesian Statistics; High Energy Astrophysics.

Teaching and Residential Assistant, Summer Science Program (seven weeks) 2019

University of Colorado Boulder

Teaching Assistant, Pomona College 2016–2018

Techniques in Observational Astronomy ($\times 2$); Spacetime, Quanta, and Entropy;

Intro to Stars and Galaxies; General Physics Lab.

Selected Talks

Conference Talks

Plenary talk, High-Resolution Exoplanet and Stellar Characterization, Granada, Spain	2026
Plenary talk, AASTCS 11: Exoplanet Atmospheres 2026, Denver, CO	2026
Plenary talk, Spirit of Lyot 6, Pasadena, CA	2026
Contributed talk, Exoplanets 6, Porto, Portugal	2026
Invited talk, Gas Giant Planet Formation: Clues from Composition Workshop, Ann Arbor, MI	2026
Plenary talk, Aspen Center for Physics, Aspen, CO	2025
Plenary talk, Cool Stars 22, San Diego, CA	2024
Plenary talk, Exoplanets 5, Leiden, Netherlands	2024
Contributed talk, ExSoCal, Pasadena, CA	2023
Contributed talk, Keck Science Meeting, Berkeley, CA	2023
Plenary talk, Spirit of Lyot 5, Leiden, Netherlands	2022

Invited Colloquia and Seminars

Königstuhl Colloquium, Max Planck Institute for Astronomy, Heidelberg, Germany	2026
Yale University Exoplanet Symposium, New Haven, CT	2026
UT Austin Astronomy Colloquium, Austin, TX	2026
Herzberg Astronomy and Astrophysics Colloquium, Victoria, Canada	2025
UCSD/SDSU Astronomy & Astrophysics Colloquium, San Diego, CA	2024
Exoplanets, Star & Planet Formation Seminar, STScI, Baltimore, MD	2026
Caltech/IPAC Lunch Seminar, Pasadena, CA	2026
CIPS Seminar, UC Berkeley, Berkeley, CA	2025
Astronomy Seminar, Carnegie EPL, Washington, DC	2025
Astrocoffee, Institute for Astronomy, Honolulu, HI	2024
Astrophysics Seminar, American Museum of Natural History, New York, NY	2022

Invited Workshops

Bridging Exoplanet and Solar System Science, Keck Institute for Space Studies, Pasadena, CA	2026
Cloud Zwei Con, Ringberg Castle, Bavaria, Germany	2023

Selected Media Coverage

How big can a planet be? With very large gas giants, it can be hard to tell. Phys.org	2026
Hydrogen sulfide detected in distant gas giant exoplanets for the first time. UCLA	2026
A brown dwarf discovered 30 years ago is actually twins circling each other. AP News	2024
Three decades later, first brown dwarf ever found offers a surprise. Reuters	2024
The first known brown dwarf is actually twins. Astronomy Magazine	2024
First Brown Dwarf Discovered is Actually Twins. AMNH	2024

Summary of Refereed Publications

Papers: 81 total · **First author:** 13 · **Student-led (mentored):** 5 · **Second-/third-author:** 23
Citations: 1,600 · ***h*-index:** 23 · *(SciX, October 2026)* · ORCID

First-Author Papers

- **Xuan**, **J. W.**, Balmer, W. O., Chachan, Y., Ruffio, J.-B., et al. 2026, "A Sulfur-rich Atmosphere for the Young Jupiter Analog AF Lep b Reveals Significant Solid Accretion", *AJ* 172, 233.
- **Xuan**, **J. W.**, Chachan, Y., Ruffio, J.-B., Ohno, K., et al. 2026, submitted.
- **Xuan**, **J. W.**, Ruffio, J.-B., Chachan, Y., Ohno, K., et al. 2026, "The Compositions of the HR 8799 Planets Reflect Accretion of Both Solids and Metal-enriched Gas", *ApJ* 1000, 27.
- Ruffio, J.-B., **Xuan**, **J. W.** (**shared first authorship**), Chachan, Y., Kesseli, A., et al. 2026, "Jupiter-like uniform metal enrichment in a system of multiple giant exoplanets", *Nature Astronomy* 10, 511–521.
- **Xuan**, **J. W.**, Perrin, M. D., Mawet, D., Knutson, H. A., et al. 2024, "Atmospheric Abundances and Bulk Properties of the Binary Brown Dwarf Gliese 229Bab from JWST/MIRI Spectroscopy", *ApJL* 977, L32.
- **Xuan**, **J. W.**, Mérand, A., Thompson, W., Zhang, Y., et al. 2024, "The cool brown dwarf Gliese 229 B is a close binary", *Nature* 634, 1070–1074.
- **Xuan**, **J. W.**, Hsu, C.-C., Finnerty, L., Wang, J., et al. 2024, "Are These Planets or Brown Dwarfs? Broadly Solar Compositions from High-resolution Atmospheric Retrievals of 10-30 MJup Companions", *ApJ* 970, 71.
- **Xuan**, **J. W.**, Wang, J., Finnerty, L., Horstman, K., et al. 2024, "Validation of Elemental and Isotopic Abundances in Late-M Spectral Types with the Benchmark HIP 55507 AB System", *ApJ* 962, 10.
- **Xuan**, **J. W.**, Wang, J., Ruffio, J.-B., Knutson, H., et al. 2022, "A Clear View of a Cloudy Brown Dwarf Companion from High-resolution Spectroscopy", *ApJ* 937, 54.
- **Xuan**, **J. W.**, Kennedy, G. M., Wyatt, M. C., Yelverton, B., 2020, "Mutual inclinations between giant planets and their debris discs in HD 113337 and HD 38529", *MNRAS* 499, 5059–5074.
- **Xuan**, **J. W.**, Wyatt, M. C., 2020, "Evidence for a high mutual inclination between the cold Jupiter and transiting super Earth orbiting π Men", *MNRAS* 497, 2096–2118.
- **Xuan**, **J. W.**, Bryan, M. L., Knutson, H. A., Bowler, B. P., et al. 2020, "A Rotation Rate for the Planetary-mass Companion DH Tau b", *AJ* 159, 97.
- **Xuan**, **W. J.**, Mawet, D., Ngo, H., Ruane, G., et al. 2018, "Characterizing the Performance of the NIRC2 Vortex Coronagraph at W. M. Keck Observatory", *AJ* 156, 156.

Student-Led

- Morgan, M., **Xuan**, **J. W.**, Bowler, B. P., Franson, K., et al. 2026, "Tracing the Influence of Distant companions on Exoplanetary Systems (TIDES) I. Highly Eccentric and Misaligned Binary Companions to Hot Jupiters", *AJ* (accepted), arXiv:2610.17811.
- Sanghi, A., Thompson, W., Mang, J., **Xuan**, **J. W.**, et al. 2026, "Worlds Next Door. IV. Mapping the Late Stages of Giant Planet Evolution with a Precise Dynamical Mass and Luminosity for ϵ Ind Ab", *AJ* 172, 115.

- Liu, Y., Zhang, Y., **Xuan**, **J. W.**, Mawet, D., et al. 2026, "Chemistry and Isotope Ratios of Substellar Atmospheres in the β Pictoris Young Moving Group and Vicinity", *ApJ* 1004, 172.
- Wang, G., **Xuan**, **J. W.**, González Picos, D., Zhang, Z., et al. 2026, "Chemical and Isotopic Homogeneity between the L Dwarf CD-35 2722 B and Its Early M Host Star", *ApJ* 997, 195.
- Sanghi, A., **Xuan**, **J. W.**, Wang, J. J., Mawet, D., et al. 2024, "Efficiently Searching for Close-in Companions Around Young M Dwarfs Using a Multiyear PSF Library", *AJ* 168, 215.

Second or Third-Author

- Kiman, R., Bowler, B. P., **Xuan**, **J. W.**, Mayer, N., et al. 2026, "Atmospheric Diversity of Giant Planet Analogs from 3-5um with SPHEREx", *arXiv e-prints*, arXiv:2608.08908.
- Thompson, W., Blakely, D., **Xuan**, **J. W.**, Blouin, S., et al. 2026, "Detecting and Characterizing Companions with a Calibrated Gaia DR2, DR3, and Hipparcos Catalog (G23H)", *AJ* 172, 53.
- Uyama, T., Currie, T., **Xuan**, **J. W.**, De Rosa, R., et al. 2026, "Discovery of a Low-mass Companion to the Accelerating Star HIP 53005 with Strongly Conflicting Mass Estimates", *AJ* 171, 332.
- Finnerty, L., Fitzgerald, M. P., **Xuan**, **J. W.**, Echeverri, D., et al. 2026, "Possible Stratospheric Emission in the Warm Neptune GJ 436 b from High-resolution Spectroscopy", *AJ* 171, 226.
- Hsu, C.-C., Wang, J. J., **Xuan**, **J. W.**, Zhang, Y., et al. 2026, "Distinct Rotational Evolution of Giant Planets and Brown Dwarf Companions", *AJ* 171, 224.
- Finnerty, L., Xin, Y., **Xuan**, **J. W.**, Inglis, J., et al. 2025, "Water Dissociation and Rotational Broadening in the Atmosphere of KELT-20 b from High-resolution Spectroscopy", *AJ* 169, 333.
- Thompson, W., Blakely, D., **Xuan**, **J. W.**, Bouchard-Côté, A., et al. 2025, "On the Orbit of the Binary Brown Dwarf Companion GL229 Ba and Bb", *AJ* 169, 193.
- Finnerty, L., Xin, Y., **Xuan**, **J. W.**, Inglis, J., et al. 2025, "True Mass and Atmospheric Composition of the Nontransiting Hot Jupiter HD 143105 b", *AJ* 169, 94.
- Hejazi, N., **Xuan**, **J. W.**, Coria, D. R., Sawczynec, E., et al. 2025, "Chemical Links between a Young M-type T Tauri Star and Its Substellar Companion: Spectral Analysis and C/O Measurement of DH Tau A", *ApJ* 978, 42.
- Zhang, Y., **Xuan**, **J. W.**, Mawet, D., Wang, J. J., et al. 2024, "Atmospheric Characterization of the Super-Jupiter HIP 99770 b with KPIC", *AJ* 168, 131.
- Hsu, C.-C., Wang, J. J., **Xuan**, **J. W.**, Ruffio, J.-B., et al. 2024, "Rotation and Abundances of the Benchmark Brown Dwarf HD 33632 Ab from Keck/KPIC High-resolution Spectroscopy", *ApJ* 971, 9.
- Costes, J. C., **Xuan**, **J. W.**, Vigan, A., Wang, J., et al. 2024, "Fresh view of the hot brown dwarf HD 984 B through high-resolution spectroscopy", *A&A* 686, A294.
- Inglis, J., Wallack, N. L., **Xuan**, **J. W.**, Knutson, H. A., et al. 2024, "Atmospheric Retrievals of the Young Giant Planet ROXs 42B b from Low- and High-resolution Spectroscopy", *AJ* 167, 218.
- Echeverri, D., **Xuan**, **J. W.**, Monnier, J. D., Delorme, J.-R., et al. 2024, "Vortex Fiber Nulling for Exoplanet Observations: First Direct Detection of M Dwarf Companions around HIP 21543, HIP 94666, and HIP 50319", *ApJL* 965, L15.
- Finnerty, L., **Xuan**, **J. W.**, Xin, Y., Liberman, J., et al. 2024, "Atmospheric Metallicity and C/O of HD 189733 b from High-resolution Spectroscopy", *AJ* 167, 43.

- Echeverri, D., **Xuan**, **J. W.**, Jovanovic, N., Ruane, G., et al. 2023, "Vortex fiber nulling for exoplanet observations: implementation and first light", *Journal of Astronomical Telescopes, Instruments, and Systems* 9, 035002.
- Xin, Y., **Xuan**, **J. W.**, Mawet, D., Wang, J., et al. 2023, "On-sky speckle nulling through a single-mode fiber with the Keck Planet Imager and Characterizer", *Journal of Astronomical Telescopes, Instruments, and Systems* 9, 035001.
- Mawet, D., Ruane, G., **Xuan**, **J. W.**, Echeverri, D., et al. 2017, "Observing Exoplanets with High-dispersion Coronagraphy. II. Demonstration of an Active Single-mode Fiber Injection Unit", *ApJ* 838, 92.

Other Refereed Publications

- Baburaj, A., Konopacky, Q. M., Theissen, C. A., **Xuan**, **J. W.**, et al. 2026, "A High-resolution Spectroscopic Survey of Directly Imaged Companion Hosts. III. Characterization of the Cold Imaged Planet Hosts AF Lep A and ϵ Indi A", *AJ* 172, 227.
- Do Ó, C. R., Thompson, W., Blakely, D., Sanghi, A., et al. 2026, "Astrometric Evidence for Outer Giant Planets in Known Exoplanet Systems", *arXiv e-prints*, arXiv:2609.37160.
- Hoch, K. K. W., Madurowicz, A., Nasedkin, E., Rickman, E., et al. 2026, "The Dyn-Atmo Survey: High-contrast Imaging Spectroscopy of the Substellar Companion HD 13724 B with the JWST NIRSpec IFU", *arXiv e-prints*, arXiv:2609.11827.
- Madurowicz, A., Rickman, E., Bardalez Gagliuffi, D., Mang, J., et al. 2026, "The Dyn-Atmo Survey: JWST/NIRSpec spectroscopy of dynamical benchmark GJ 758 B", *arXiv e-prints*, arXiv:2609.11828.
- Sanghi, A., Quarles, B., Lin, A. S. J., Shariat, C., et al. 2026, "Worlds Next Door. V. A Candidate Solar System Scale Super-Jupiter in the 61 Cygni Binary System", *arXiv e-prints*, arXiv:2609.32893.
- Shariat, C., El-Badry, K., Wang, S., Wang, X.-Y., et al. 2026, "Stellar companions sculpt hot Jupiter formation and spin-orbit evolution", *arXiv e-prints*, arXiv:2609.19249.
- Finnerty, L., Fitzgerald, M. P., Xin, Y., **Xuan**, **J. W.**, et al. 2026, "Atmospheric Characterization of Six Ultrahot Jupiters from K-band High-resolution Spectroscopy", *AJ* 172, 119.
- Van Zandt, J., Bowler, B. P., Endl, M., Cochran, W. D., et al. 2026, "The γ Cephei System: Updated Orbits, Dynamical Architecture, and Limits on Additional Companions", *arXiv e-prints*, arXiv:2608.30030.
- Van Zandt, J., Bowler, B. P., Petigura, E. A., Franson, K., et al. 2026, "Gaia Exoplanet Orbits, Demographics, and Evolution Survey (GEODES). I. Characteristics of Three Long-period Companions Accelerating their Host Stars", *ApJ* 1006, 222.
- Baburaj, A., Ruffio, J.-B., Perrin, M., **Xuan**, **J. W.**, et al. 2026, "JWST-TST High Contrast: First Direct Spectroscopy of GJ 504 b Reveals Clouds and Possible Metal Enrichment", *AJ* 172, 28.
- Gibbs, A., Ruffio, J.-B., Bidot, A., Barman, T. S., et al. 2026, "Discovery of an Exterior Third Planet Orbiting β Pictoris", *ApJ* 1006, L11.
- von Stauffenberg, A., Sauter, J., Mollière, P., Ravet, M., et al. 2026, "^{13}CO and potential variability in β Pictoris b with GRAVITY+", *A&A* 711, L2.
- Zhang, Y., Huang, C., Householder, A., Owen, J. E., et al. 2026, "Hydrogen Airglow from an Escaping Ultrahot Jupiter Atmosphere", *ApJ* 1005, L29.
- Inglis, J., Millar-Blanchaer, M. A., Konopacky, Q., **Xuan**, **J. W.**, et al. 2026, "The Orbital Architecture and Origins of the Young, Planetary-mass Companion Host System, ROXs 42B", *AJ* 171, 280.

- Whitebook, S., Rodriguez, A. C., Burdge, K., Prince, T., et al. 2026, "A Mass Transferring Brown Dwarf Binary on a 57 Minute Orbit", *ApJ* 1000, L23.
- Blakely, D., Thompson, W., Johnstone, D., Speedie, J., et al. 2026, "Dynamical Mass Constraints on Transition Disk Perturbers with the G23H Catalog", *AJ* (accepted).
- Zhang, J., Huber, D., Bottom, M., Weiss, L. M., et al. 2026, "Dynamical Architectures of S-type Transiting Planets in Binaries. II. A Dichotomy in Orbital Alignment of Small Planets in Close Binary Systems", *AJ* 171, 77.
- Zhang, Y., Wardenier, J. P., Householder, A., Komacek, T. D., et al. 2026, "Extreme Winds on the Emerging Dayside of an Ultrahot Jupiter", *ApJL* 997, L40.
- Jones, N. K., Wang, J. J., Nielsen, E. L., De Rosa, R. J., et al. 2025, "HD 143811 AB b: A Directly Imaged Planet Orbiting a Spectroscopic Binary in Sco-Cen", *ApJL* 995, L41.
- Peck, A. E., Roberson, W., Nielsen, E. L., De Rosa, R. J., et al. 2025, "Characterization of the Host Binary of the Directly Imaged Exoplanet HD 143811 AB b", *ApJL* 995, L40.
- Finnerty, L., Inglis, J., Fitzgerald, M. P., Echeverri, D., et al. 2025, "The Watery Atmosphere of HD 209458 b Revealed by Joint K- and L-band High-resolution Spectroscopy", *AJ* 170, 223.
- Uyama, T., Kuzuhara, M., Beichman, C., Hirano, T., et al. 2025, "Direct Imaging Explorations for Companions from the Subaru/IRD Strategic Program II; Discovery of a Brown-dwarf Companion around a Nearby Mid-M-dwarf LSPM J1446+4633", *AJ* 170, 272.
- Horstman, K. A., Ruffio, J.-B., Wang, J. J., Hsu, C.-C., et al. 2025, "Fringing analysis and forward modeling of Keck Planet Imager and Characterizer (KPIC) spectra", *Journal of Astronomical Telescopes, Instruments, and Systems* 11, 035004.
- Zhang, J., Weiss, L. M., Huber, D., **Xuan**, **J. W.**, et al. 2025, "Discovery of a Jupiter Analog Misaligned to the Inner Planetary System in HD 73344", *AJ* 169, 200.
- Sappey, B., Konopacky, Q., Ó, C. R. D., Barman, T., et al. 2025, "HD 206893 B at High Spectral Resolution with the Keck Planet Imager and Characterizer", *AJ* 169, 175.
- Jovanovic, N., Echeverri, D., Delorme, J.-R., Finnerty, L., et al. 2025, "Technical description and performance of the phase II version of the Keck Planet Imager and Characterizer", *Journal of Astronomical Telescopes, Instruments, and Systems* 11, 015005.
- Hsu, C.-C., Wang, J. J., Blake, G. A., **Xuan**, **J. W.**, et al. 2024, "PDS 70b Shows Stellar-like Carbon-to-oxygen Ratio", *ApJL* 977, L47.
- Zhang, J., Huber, D., Weiss, L. M., **Xuan**, **J. W.**, et al. 2024, "A Testbed for Tidal Migration: The 3D Architecture of an Eccentric Hot Jupiter HD 118203 b Accompanied by a Possibly Aligned Outer Giant Planet", *AJ* 168, 295.
- Zhang, Y., González Picos, D., de Regt, S., Snellen, I. A. G., et al. 2024, "The ESO SupJup Survey. III. Confirmation of ^{13}CO in YSES 1 b and Atmospheric Detection of YSES 1 c with CRIRES^{+}", *AJ* 168, 246.
- Horstman, K., Ruffio, J.-B., Batygin, K., Mawet, D., et al. 2024, "RV Measurements of Directly Imaged Brown Dwarf GQ Lup B to Search for Exosatellites", *AJ* 168, 175.
- Morris, E. C., Wang, J. J., Hsu, C.-C., Ruffio, J.-B., et al. 2024, "κ Andromedae b Is a Fast Rotator from KPIC High-resolution Spectroscopy", *AJ* 168, 144.
- Wallack, N. L., Ruffio, J.-B., Ruane, G., Ren, B. B., et al. 2024, "A Survey of Protoplanetary Disks Using the Keck/NIRC2 Vortex Coronagraph", *AJ* 168, 78.

- Do Ó, C. R., Sappey, B., Konopacky, Q. M., Ruffio, J.-B., et al. 2024, "Orbital and Atmospheric Characterization of the 1RXS J034231.8+121622 System using High-resolution Spectroscopy Confirms that the Companion is a Low-mass Star", *AJ* 167, 278.
- Salama, M., Guthery, C., Chambouleyron, V., Jensen-Clem, R., et al. 2024, "Keck Primary Mirror Closed-loop Segment Control Using a Vector-Zernike Wavefront Sensor", *ApJ* 967, 171.
- Maire, A.-L., Delrez, L., Pozuelos, F. J., Becker, J., et al. 2023, "Workshop Summary: Exoplanet Orbits and Dynamics", *PASP* 135, 106001.
- Finnerty, L., Schofield, T., Sappey, B., **Xuan**, **J. W.**, et al. 2023, "Keck Planet Imager and Characterizer Emission Spectroscopy of WASP-33b", *AJ* 166, 31.
- Long, F., Ren, B. B., Wallack, N. L., Harsono, D., et al. 2023, "A Large Double-ring Disk Around the Taurus M Dwarf J04124068+2438157", *ApJ* 949, 27.
- Ruffio, J.-B., Horstman, K., Mawet, D., Rosenthal, L. J., et al. 2023, "Detecting Exomoons from Radial Velocity Measurements of Self-luminous Planets: Application to Observations of HR 7672 B and Future Prospects", *AJ* 165, 113.
- Wang, J., Wang, J. J., Ruffio, J.-B., Blake, G. A., et al. 2023, "Retrieving C and O Abundance of HR 8799 c by Combining High- and Low-resolution Data", *AJ* 165, 4.
- Wang, J., Kolecki, J. R., Ruffio, J.-B., Wang, J. J., et al. 2022, "Retrieving the C and O Abundances of HR 7672 AB: A Solar-type Primary Star with a Benchmark Brown Dwarf", *AJ* 163, 189.
- Quiroz, J., Wallack, N. L., Ren, B., Dong, R., et al. 2022, "Improving Planet Detection with Disk Modeling: Keck/NIRC2 Imaging of the HD 34282 Single-armed Protoplanetary Disk", *ApJL* 924, L4.
- Llop-Sayson, J., Wang, J. J., Ruffio, J.-B., Mawet, D., et al. 2021, "Constraining the Orbit and Mass of epsilon Eridani b with Radial Velocities, Hipparcos IAD-Gaia DR2 Astrometry, and Multiepoch Vortex Coronagraphy Upper Limits", *AJ* 162, 181.
- Wang, J. J., Ruffio, J.-B., Morris, E., Delorme, J.-R., et al. 2021, "Detection and Bulk Properties of the HR 8799 Planets with High-resolution Spectroscopy", *AJ* 162, 148.
- Bryan, M. L., Ginzburg, S., Chiang, E., Morley, C., et al. 2020, "As the Worlds Turn: Constraining Spin Evolution in the Planetary-mass Regime", *ApJ* 905, 37.
- Ruane, G., Ngo, H., Mawet, D., Absil, O., et al. 2019, "Reference Star Differential Imaging of Close-in Companions and Circumstellar Disks with the NIRC2 Vortex Coronagraph at the W. M. Keck Observatory", *AJ* 157, 118.